

Problem #	Query	Result
1	<pre>-- Problem 1 1 select power_plants.powerplantname as 'Plant Name', 2 countries.countryname as "Country", operators.operatorname as 'Op Name', 3 fuel_types.Category, fuel_types.fuelname, 4 power_plants.CapacityMW, power_plants.CommissionYear 5 from power_plants 6 join countries on countries.countrycode = power_plants.countrycode 7 join operators on operators.operatorid = power_plants.operatorid 8 join fuel_types on fuel_types.fuelid = power_plants.fuelid 9 order by CapacityMW desc;</pre>	<pre>1 +-----+-----+-----+-----+-----+-----+-----+ 2 Plant Name Country Op Name Category fuelname CapacityMW CommissionYear 3 +-----+-----+-----+-----+-----+-----+-----+ 4 Three Gorges China China Yangtze Power Renewable Hydro 22500.00 2003 5 Itaipu Brazil Electrobras Renewable Hydro 14000.00 1984 6 Tuoketuo China China Yangtze Power Fossil Coal 13200.00 1995 7 Kashiwazaki-Kariwa Japan Tepco Energy Nuclear Uranium 8212.00 1985 8 Grand Coulee United States Global Hydro Inc Renewable Hydro 6809.00 1942 9 Vindhyachal India NTPC Limited Fossil Coal 4760.00 1987 10 Janschwalde Germany Rhein Power AG Fossil Coal 3000.00 1981 11 Loy Yang Australia Snowy Hydro Fossil Coal 2210.00 1984 12 +-----+-----+-----+-----+-----+-----+-----+</pre>
2	<pre>-- Problem 2 1 select power_plants.powerplantname as 'Plant Name', power_plants.countrycode, 2 generation_records.Year, generation_records.GenGWh from power_plants 3 join generation_records on generation_records.powerplantid = power_plants.powerplantid 4 where generation_records.year = 2024 5 order by GenGWh desc;</pre>	<pre>1 +-----+-----+-----+-----+-----+ 2 Plant Name countrycode Year GenGWh 3 +-----+-----+-----+-----+-----+ 4 Three Gorges CHN 2024 98800.00 5 Itaipu BRA 2024 89000.10 6 Tuoketuo CHN 2024 71000.30 7 Vindhyachal IND 2024 35600.80 8 Grand Coulee USA 2024 21500.50 9 Janschwalde DEU 2024 18400.20 10 Loy Yang AUS 2024 14200.40 11 Kashiwazaki-Kariwa JPN 2024 0.00 12 +-----+-----+-----+-----+-----+</pre>
3	<pre>-- Problem 3 1 select power_plants.powerplantname as 'Plant Name', countries.countrycode, generation_records.Year, 2 GenGWh, CO2EmissionsTonnes from power_plants 3 join countries on countries.countrycode = power_plants.countrycode 4 join generation_records on generation_records.powerplantid = power_plants.powerplantid 5 join emission_metrics on emission_metrics.powerplantid = power_plants.powerplantid 6 where generation_records.Year = 2024 7 order by CO2EmissionsTonnes asc;</pre>	<pre>1 +-----+-----+-----+-----+-----+-----+ 2 Plant Name countrycode Year GenGWh CO2EmissionsTonnes 3 +-----+-----+-----+-----+-----+-----+ 4 Grand Coulee USA 2024 21500.50 0 5 Itaipu BRA 2024 89000.10 0 6 Kashiwazaki-Kariwa JPN 2024 0.00 0 7 Three Gorges CHN 2024 98800.00 0 8 Loy Yang AUS 2024 14200.40 1950000 9 Janschwalde DEU 2024 18400.20 2450000 10 Vindhyachal IND 2024 35600.80 3100000 11 Tuoketuo CHN 2024 71000.30 5200000 12 +-----+-----+-----+-----+-----+-----+ 13 -- Total CO2 = 5200000 + 3100000 + 2450000 + 1950000 = 12700000</pre>

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4
-- write 4
with total_cum_power_gen as (
  select operators.operatorname, HQCountry, sum(Generation) as Gen
  from operators
  left join power_plants on power_plants.operatorid = operators.operatorid
  left join generation_records on generation_records.powerplantid = power_plants.powerplantid
  group by operators.operatorname, operators.HQCountry
)
select * from total_cum_power_gen
order by gen desc; -- only seven rows because one Operator isn't in power_plants
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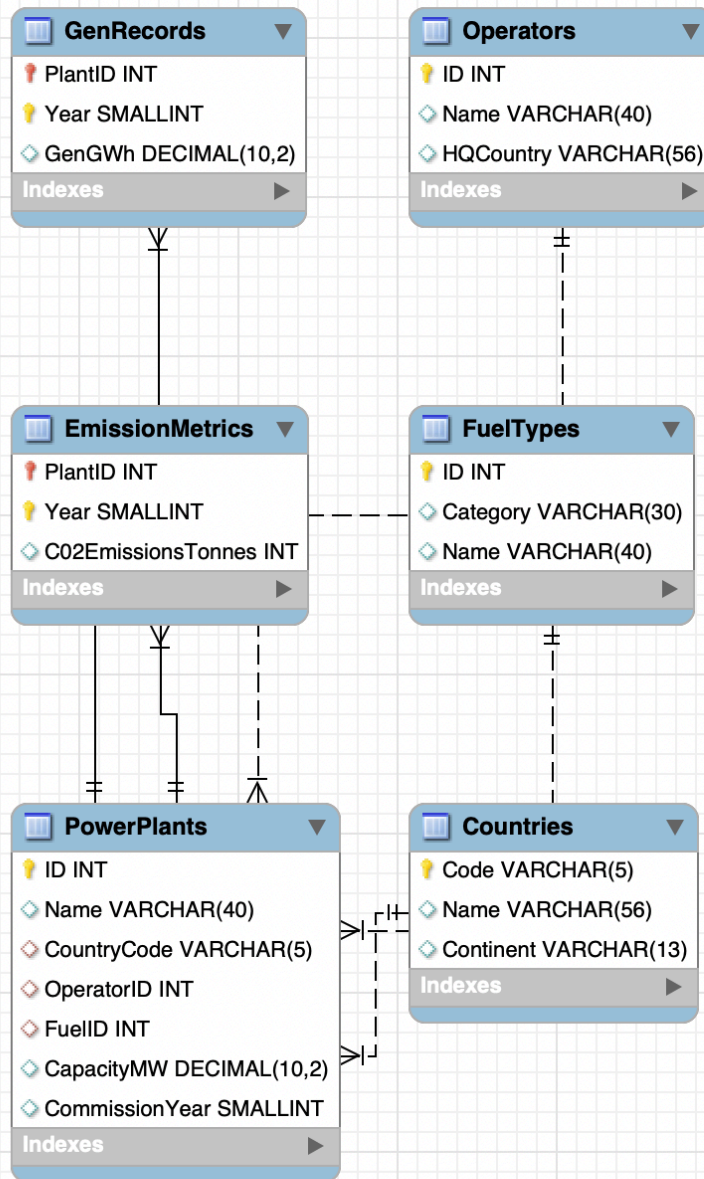
4.sql

operatorname	HQCountry	Gen
China Yangtze Power	CHN	169800.30
Electrobras	BRA	89000.10
NTPC Limited	IND	35600.80
Global Hydro Inc	USA	21500.50
Rhein Power AG	DEU	18400.20
Snowy Hydro	AUS	14200.40
Tepco Energy	JPN	0.00
Eskom Holdings	ZAF	NULL

```
5
-- write 5
with total_gen as (
  select countries.countrycode, sum(Generation) as Generation
  from countries
  join power_plants on countries.countrycode = power_plants.countrycode
  join generation_records on generation_records.powerplantid = power_plants.powerplantid
  group by countries.countrycode
),
total_emissions as (
  select countries.countrycode, sum(CO2EmissionsTonnes) as Emissions
  from countries
  join power_plants on countries.countrycode = power_plants.countrycode
  join emission_metrics on emission_metrics.powerplantid = power_plants.powerplantid
  group by countries.countrycode
)
select countries.countryname, total_gen.generation, total_emissions.emissions
from countries
join total_emissions on total_emissions.countrycode = countries.countrycode
join total_gen on total_gen.countrycode = countries.countrycode
order by generation desc;
```

5.sql

countryname	generation	emissions
China	169800.30	5200000
Brazil	89000.10	0
India	35600.80	3100000
United States	21500.50	0
Germany	18400.20	2450000
Australia	14200.40	1950000
Japan	0.00	0



EER Diagram: